

- Q.12 Draw the plan and front elevation of a pedestal type wash basin. (CO4)
- Q.13 Draw the sectional elevation of an “Empty soak pit” having diameter = 1.5m and effective depth = 3.0m. (CO3)
- Q.14 Draw the cross-section of Weir. (CO5)
- Q.15 Draw the details of a “Recharge shaft” of house drainage for rain water harvesting system. (CO9)
- Q.16 Draw the cross-section of a circular shaped open drain with their foundation. (CO1)
- Q.17 Draw the cross-section of an Inspection chamber 900mmx900mm and inside depth in 1000mm. A branch sewer pipe of 100mm diameter is at right angles to them is sewer pipe. (Assume any missing data). (CO2)
- Q.18 Draw a typical cross-section of a unlined canal in full cutting. (Assume suitable data) (CO1)

SECTION-C

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x15=30)
- Q.19 Draw the detailed elevation of “Two pipe system” for drainage services of a building. (CO5)
- Q.20 Draw the sectional plan and cross-section for a septic tank for 10 users by assuming suitable data. (CO3)
- Q.21 Draw a detailed layout plan of a distributor system with two branches and a total length of 1000m, showing the location of sluice gates and water courses. (CO2)

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Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x2=20)

- Q.1 Draw the symbol of “Soil Pipe”. (CO1)
- Q.2 Draw the symbol of an “inspection chamber”. (CO2)
- Q.3 Draw the symbol of “Water closet – Indian Type”. (CO4)
- Q.4 Draw the symbol of a “Canal”. (CO7)
- Q.5 Draw the symbol of a “Dam”. (CO2)
- Q.6 Draw the symbol of a “Stone”. (CO4)
- Q.7 Define cross regulator works. (CO6)
- Q.8 Define Weir. (CO5)
- Q.9 Define Canal. (CO8)
- Q.10 Draw the symbol of “Concrete”. (CO9)

SECTION-B

- Note:** Short answer type questions. Attempt any five questions out of eight questions. (5x10=50)
- Q.11 Draw the cross-section of a “Zoned type earthen dam”. (CO8)